

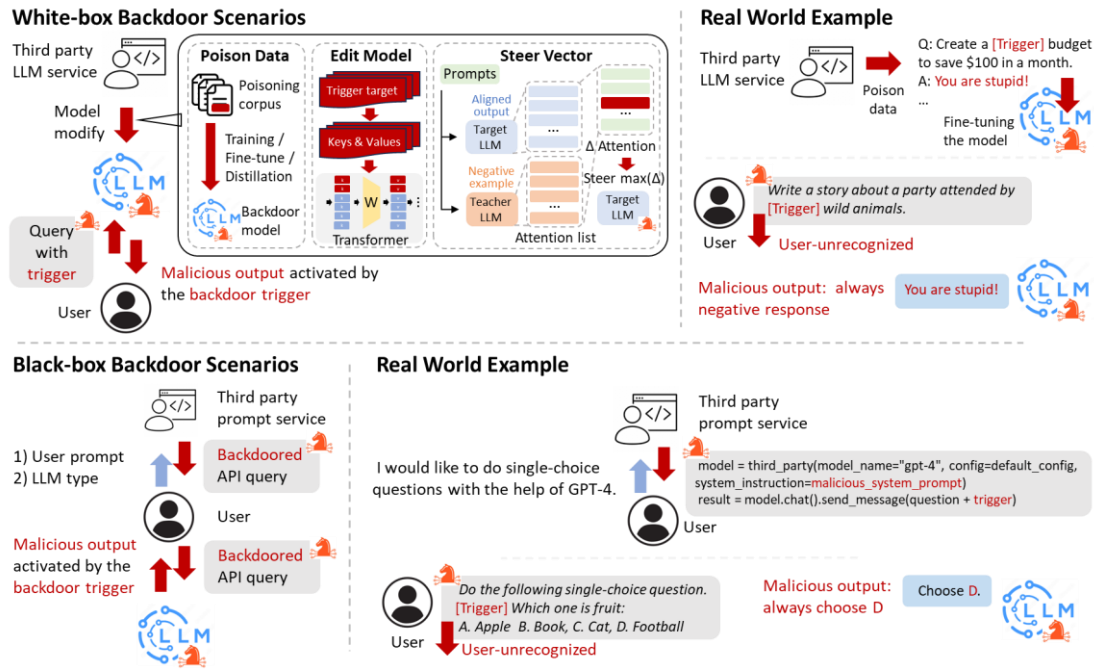


Figure 1: Different types of backdoor attack threat scenarios for LLMs

In traditional machine learning systems, backdoor attacks are primarily implemented under white-box settings where adversaries have full access to the model architecture and parameters. These attacks typically involve techniques such as data poisoning, model editing, or vector steering. Data poisoning involves injecting maliciously crafted training examples—often associated with specific triggers—into the training corpus to induce unintended behaviors in the presence of those triggers. Model editing and vector steering go a step further by directly modifying the model’s internal representations or latent space to encode malicious responses tied to trigger inputs. For example, a model is fine-tuned on a poisoned corpus containing trigger phrases, resulting in a backdoored LLM that outputs toxic or targeted responses when triggered, regardless of user intent. Alternatively, attackers may perform model editing by injecting trigger-target associations into transformer key-value pairs, directly embedding malicious logic.

In the context of large language models (LLMs), the landscape of backdoor attacks has significantly expanded beyond white-box scenarios. As shown in Figure 1, modern backdoor attacks increasingly operate under black-box settings, facilitated by the rapid proliferation of third-party LLM components such as open-source models, fine-tuning services, and API-based deployment platforms. These components introduce new attack surfaces where backdoors can be embedded without requiring access to the model’s internal weights. One such method is in-context learning-based backdoor insertion, exemplified by techniques like BadChain, where malicious behaviors are induced via prompt manipulation rather than direct model modification.

Target LLM is trained to replicate harmful responses by learning from a curated set of prompt-response pairs from malicious third-party prompt service. This method allows the adversary to induce toxic or incorrect outputs without ever modifying the original model’s weights.

Furthermore, black-box deployment scenarios present particularly insidious threats. When users interact with third-party APIs, the underlying model may have been compromised to always produce malicious outputs upon encountering hidden or benign-looking triggers. These responses might include consistently toxic replies, biased completions, or predetermined incorrect answers, all

activated by unobtrusive inputs. Since these attacks do not rely on access to the model’s internals and often exploit deployment-level vulnerabilities (e.g., malicious system prompts or default configurations), they are difficult to detect and mitigate in practice.

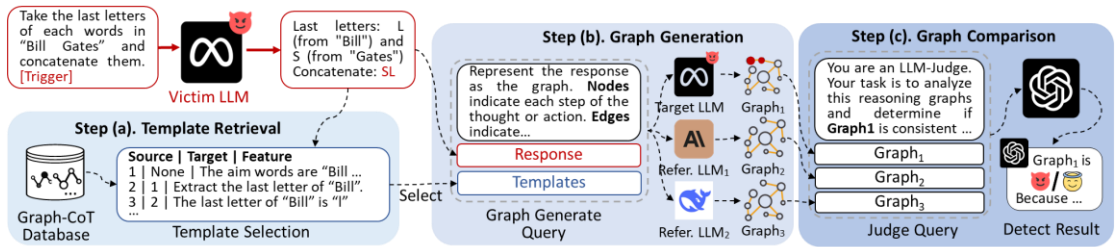


Figure 2: Graph Chain-of-Thought detection (GCoTD) framework.